

SHIVAM SINGH

AI Researcher | Vision-Language Models | Multimodal Reasoning | Embodied AI

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SUMMARY

AI Researcher with **3+ years** of research experience working on Vision-Language Models, multimodal reasoning, video understanding, and embodied AI. Developed multimodal foundation models and video reasoning systems, built scalable multimodal data pipelines, and worked on LLM-based planning, task anticipation, and human-robot collaboration.

EXPERIENCE

pixelstwo

Collaboration

Sep 2026 – Present

Remote

Focus: Video Data Quality & Real-to-Sim / Sim-to-Real Systems

- **Quality control of video data** for reliable, consistent downstream training and evaluation; working on **real-to-sim and sim-to-real systems**.

intuition.dev

Collaboration

Aug 2026 – Sep 2026

Remote

Focus: Real-World Video Pipelines

- **Label-free Segmentation and Grounded Description of Egocentric Work Videos** 2026
 - Built a **label-free, geometry-first pipeline** that splits raw body-camera videos into **clean sub-task clips** using classical detectors and camera-trajectory **task-transition detection**, with no large models or prior annotations.
 - Isolated task segments before VLM captioning, **reducing VLM inference cost** and producing consistent, grounded annotations under a fixed JSON schema. [Technical Report]

AMD India Private Limited

Research Fellow

Jun 2025 – Jul 2026

AI Models & Agents Team

Focus: VLMs, Video Reasoning & Spatial Understanding

Supervisors: Saptarshi Majumder, Pratik Prabhanjan Brahma

- **Spatial Video Understanding from Raw RGB Videos** 2026
 - Proposed *SlotMem*, a **memory-augmented** transformer for object tracking, spatial understanding, and long-horizon reasoning from raw monocular RGB video; achieved **state-of-the-art** on OST-Bench (**44.4%**).
 - Built a scalable supervision pipeline (Depth-Anything-3, YOLO-World, VLMs) that reconstructs 3D scenes and camera trajectories to generate spatial, temporal, and goal-oriented reasoning data.
- **Pause and Think: Video-Grounded Assistive Action Suggestion** IROS 2026
 - **First author.** Introduced *Pause-and-Think-T* (~10k reasoning QA pairs) and *Pause-and-Think-B* (**300-sample** benchmark) for grounded, goal-oriented action suggestion across Epic-Kitchens, Assembly101, and Ego4D.
 - Fine-tuned a compact **4B** VLM on 8×AMD Instinct MI325 GPUs: **58.0%** accuracy, matching Qwen3-VL-235B (**58.9%**, **59× larger**) and GPT-5.2, outperforming GPT-4o, with strong zero-shot transfer to EgoThink and TempCompass.
 - Deployed on **AMD Ryzen AI** for real-time on-device inference. [Code | Paper]

EDUCATION

IIIT Hyderabad

M.S.

Electronics and Communication Engineering

2023 – 2026

CGPA: 8.33/10

RKGIT, Ghaziabad

B.Tech

Electronics and Communication Engineering

2017 – 2021

CGPA: 8.07/10

SKILLS

Research Areas

LLMs, VLMs, VLAs, Reasoning Models, Computer Vision, Multimodal Learning, Agentic AI, Embodied AI

Training & Inference

PyTorch, Hugging Face, LLaMA-Factory, SFT, RLHF, LoRA, DeepSpeed, vLLM, ROCm

Data & Infrastructure

Multimodal Data Curation, Synthetic Data, Benchmarking, Docker, Slurm, OpenCV, Open3D

Programming & Robotics

Python, C++, C, PDDL, RDDL, ROS, Gazebo, AI2-THOR

RESEARCH HIGHLIGHTS

- **3** first/co-first author papers at **A*** robotics venues (**ICRA 2024, ICRA 2025, IROS 2026**)
- Built a compact **4B** VLM for video-grounded spatial understanding, matching a **59×** larger model
- Achieved **SOTA** on OST-Bench (**44.4%**) through improved spatial reasoning and memory
- Built scalable **multimodal data curation pipelines**, from structuring raw real-world work videos to generating grounded VLM training data
- Trained multimodal models on **8×** AMD Instinct MI325 GPUs and deployed on-device on AMD Ryzen AI

IIIT Hyderabad

M.S. Researcher

Focus: Embodied AI & Robot Planning

Supervisor: Prof. Madhava Krishna

Collaborators: Dr. Mohan Sridharan, Dr. Krishna Murthy Jatavallabhula, Dr. Brojeshwar Bhowmick

Jan 2023 – May 2025

Robotics Research Centre

- **Hybrid Planning with LLMs & Probabilistic Models** **ECMR 2025**
Mentor: Guided framework design and ablations for LLM-based task planning under uncertainty in multi-agent settings. [Paper]
- **LLM & Knowledge Graphs for Task Decomposition** **IEEE ICRA 2025**
Project Lead: Combined LLMs, knowledge graphs, and human-in-the-loop feedback so embodied agents plan unseen long-horizon tasks and progressively adapt to new environments. [Website | Paper]
- **Task Anticipation for Multi-Agent Systems** **ICRA & RSS 2024 Workshops**
Project Lead: Extended LLM-based task anticipation to multi-agent human-robot collaboration, adapting to changing human behavior and environmental uncertainty. [Website | Paper]
- **Task Anticipation using LLMs** **IEEE ICRA 2024**
Co-author & Project Lead: Household agent that anticipates user tasks from learned behavior patterns with LLMs and plans them with PDDL for long-horizon execution. [Website | Paper]

AlsoEnergy

Technical Support Engineer

Apr 2021 – Dec 2021

India

- PowerTrack platform support for commercial solar clients: troubleshooting, data visualization, hardware integration.

PUBLICATIONS

- **Pause and Think: A Dataset and Benchmark for Video-Grounded Assistive Action Suggestion.** Shivam Singh, S Majumder, P Prabhanjan, Z Liu, E Barsoum.
IEEE/RSJ IROS 2026 (A*). Work done at AMD.
- **Anticipate, Adapt, Act: A Hybrid Framework for Task Planning.** N Dash, A Kaura, Shivam Singh, R Singh, S Banerjee, M Sridharan, M Krishna.
IEEE ECMR 2025.
- **AdaptBot: Combining LLM with Knowledge Graphs and Human Input for Generic-to-Specific Task Decomposition and Knowledge Refinement.** Shivam Singh, K Swaminathan, N Dash, R Singh, S Banerjee, M Sridharan, M Krishna.
IEEE ICRA 2025 (A*), Atlanta, US.
- **Anticipate & Collab: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration.** Shivam Singh*, K Swaminathan*, R Arora*, D Das, S Banerjee, M Sridharan, M Krishna.
RSS 2024 Workshop on Task Specification for General-Purpose Intelligent Robots, Delft, Netherlands.
- **Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments.** R Arora*, Shivam Singh*, K Swaminathan, A Datta, S Banerjee, B Bhowmick, KM Jatavallabhula, M Sridharan, M Krishna.
IEEE ICRA 2024 (A*), Yokohama, Japan.
- **DaTAPlan: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration.** Shivam Singh*, K Swaminathan*, R Arora*, R Singh, A Datta, D Das, S Banerjee, M Sridharan, M Krishna.
ICRA 2024 Workshop on Cooking Robotics: Perception and Motion Planning, Yokohama, Japan.

PATENTS

Systems and Methods for Data-driven Task Anticipation and Knowledge-driven Planning for Human-Robot Collaboration

Shivam Singh, K Swaminathan, R Arora, R Singh, A Datta, D Das, S Banerjee, M Sridharan, M Krishna.

US Patent App. 19089689

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